

Ames Housing Market Intelligence

A Data-Driven Guide for Real Estate Investment Decisions

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The Story

The Problem:

Real estate investors in Ames, Iowa face uncertainty about which home features drive value and when to buy or sell.

The Approach:

Analyzed 2,930 home sales (2006-2010) covering the housing bubble peak, financial crisis, and recovery to identify patterns that still apply today.

The Outcome:

Data-driven strategies for pricing homes accurately, timing the market, and maximizing investment returns.

Key Findings

+\$32,000

per quality point increase
(Strongest predictor: $r=0.80$)

\$20,000

seasonal swing
(June peak vs January trough)

-\$650/year

age depreciation
(Renovation doesn't overcome old age)

22% decline

during 2008 crisis
(Recovery began late 2009)

The Data

Dataset Overview

- 2,930 homes sold in Ames, Iowa
- 82 variables including size, quality, age, features, location
- Time period: **2006-2010** (housing bubble peak → financial crisis → recovery)

Why This Period Matters?

Captures a complete market cycle showing how homes hold value during both boom and bust times.

Audience

Ames Real Estate Investment Group

Sellers

When to list? Which improvements add value?

Buyers

When to shop? What features justify premium prices?

Investors

How to maximize ROI? Where is the sweet spot?

Developers

What to build? Which features optimize profit?

Finding #1: Quality Is King

Overall quality rating (1-10 scale) is the strongest predictor of home value, with a correlation of 0.80. Each quality point adds approximately \$32,000 to the sale price.

Quality Level	Average Price	Difference
5 (Average)	\$130,000	Baseline
7 (Good)	\$194,000	+\$64,000 (+49%)
9 (Excellent)	\$258,000	+\$128,000 (+98%)

 **Takeaway:** Focus upgrades on quality improvements (materials, finishes, craftsmanship) over size additions for best ROI.

Finding #2: Timing Matters

Sale prices fluctuate \$20,000 between summer peaks and winter troughs. The 2008 crisis caused a 22% decline, but recovery began in late 2009.

Seasonal Pattern:

- **June peak:** ~\$175,000 avg
- **January trough:** ~\$155,000 avg
- **\$20,000 difference** = 12% premium for summer listings

Crisis Impact:

- **2007 peak:** \$185,000 avg
- **2009 trough:** \$144,000 avg
- **22% decline** during financial crisis
- **Recovery:** Began late 2009



Takeaway: List in May-June for +\$15K premium. Buy in January-February to save \$12K. Crisis troughs offer 20-30% discounts.

The Pricing Model

Built a multiple regression model to predict home prices based on the four strongest drivers identified in hypothesis testing.

Model Performance

- $R^2 = 0.76$ (explains 76% of price variation)
- RMSE = \$39,000 (typical error)

Top Predictors:

Predictor	Impact
Overall Quality	+\$32,000 per point
Living Area (sq ft)	+\$6,000 per 100 sq ft
Garage Area (sq ft)	+\$2,500 per 100 sq ft
Age at Sale (years)	-\$650 per year

 **Takeaway: Quality upgrades deliver 5x more value than size additions. Focus on materials and finishes, not square footage.**

Actionable Recommendations



For Sellers

- List May-June for +\$15K premium
- Focus quality upgrades (highest ROI)
- Modern finishes > room additions
- Avoid major renovations on old homes



For Buyers

- Shop January-February (save \$12K)
- Don't overpay for renovated old homes
- Quality matters more than size
- Negotiate harder in winter months



For Investors

- Buy crisis troughs (20-30% discount)
- Target quality 8, ~1,900 sq ft homes
- Quality upgrades deliver 5x ROI vs size
- Avoid luxury spec (model less accurate)



For Developers

- Build quality 8, 1,900 sq ft (sweet spot)
- Premium materials > square footage
- Avoid luxury spec homes (>\$300K)
- Target June listings for best prices

Methodology

Statistical Approach:

1. Exploratory Data Analysis

- Distributions, correlations, outliers, missing data assessment
- Identified quality ($r=0.80$) and size ($r=0.71$) as top predictors

2. Hypothesis Testing

- Central Air: t-test showed +\$84K premium ($p<0.001$)
- Recent Construction: +\$92K for homes built 2000+ ($p<0.001$)
- House Style: ANOVA confirmed significant price differences ($p<0.001$)

3. Multiple Regression Model

- $R^2=0.76$, RMSE=\$39K
- Predictors: Overall Quality, Living Area, Garage Area, Age at Sale
- Diagnostics: Mild heteroscedasticity, Q-Q plot good in center

4. Time Series Analysis

- Seasonal decomposition, ACF/PACF, trend identification

Questions?

Thank you for your attention!

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